DIGITAL ELECTRONICS

Unit-1: Fundamentals of Semiconductor Physics

Comprehensive Study Notes

HPU BCA Semester-I

Topic 1: Energy Bands in Solids

1.1 Introduction to Energy Bands

The electrical conductivity of a material determines whether it behaves as an **insulator**, **semiconductor**, or **conductor**. This behavior is explained by **energy band theory**, which describes the allowed energy levels for electrons in a solid.

```
ENERGY BAND DIAGRAMS

INSULATOR          SEMICONDUCTOR       CONDUCTOR
+-----------+      +-----------+      +-----------+
|           |      |           |      |           |
|  Conduc-  |      |  Conduc-  |      |  Conduc-  |
|  tion     |      |  tion     |      |  tion &   |
|  Band     |      |  Band     |      |  Valence  |
|           |      |           |      |  Bands    |
|  +----+   |     |  +----+   |     | Overlap   |
|  |    |   |     |  |    |   |     |           |
|  |Gap|    |     |  |Gap|    |     +-----------+
|  |5eV|    |     |  |1eV|    |
|  |    |   |     |  |    |   |
|  +----+   |     |  +----+   |
|           |      |           |
|  Valence  |      |  Valence  |
|  Band     |      |  Band     |
+-----------+      +-----------+
```

Explanation of Energy Bands

Band	Description
Valence Band	The highest range of electron energies where electrons are bound to atoms and do not contribute to electric current
Conduction Band	The range of electron energies higher than the valence band where electrons are free to move under the influence of an external voltage
Forbidden Band Gap	The energy gap between the valence and conduction bands. It represents the energy required for an electron to move from the

	valence band to the conduction band

Topic 2: Classification of Materials

2.1 Classification Based on Energy Bands

Material Type	Band Gap	Conductivity	Examples
Insulator	Large (>5 eV)	Very low	Glass, Mica, Rubber
Semiconductor	Moderate (~1 eV)	Moderate (temperature-dependent)	Silicon, Germanium, GaAs
Conductor	No gap (overlap)	Very high	Copper, Aluminum, Silver

Key Points

- **Silicon (Si):** Band gap ~1.21 eV at absolute zero temperature, 1.1 eV at room temperature
- **Germanium (Ge):** Band gap ~0.785 eV at absolute zero temperature
- At absolute zero temperature (0K), semiconductors behave as insulators
- As temperature increases, electrons gain energy to cross the band gap and move to the conduction band

Topic 3: Intrinsic & Extrinsic Semiconductors

3.1 Semiconductor Types

```
SEMICONDUCTOR TYPES
+-----------------------------------------------------+
|                    SEMICONDUCTOR                     |
+-----------------------------------------------------+
         |                              |
         v                              v
+-------------------------+  +-------------------------+
|      INTRINSIC          |  |      EXTRINSIC          |
|  (Pure Semiconductor)   |  |      (Doped)            |
+-------------------------+  +-------------------------+
                                    |              |
                                    v              v
                    +------------------------+  +------------------------+
                    |   N-Type: Pentavalent  |  |  P-Type: Trivalent     |
                    |   impurity added       |  |  impurity added        |
                    |   (P, As, Sb)          |  |  (B, Ga, In)           |
                    +------------------------+  +------------------------+
```

3.2 N-Type Semiconductor

- Formed by doping intrinsic semiconductor with **pentavalent** impurities (Phosphorus, Arsenic, Antimony)
- Four out of five valence electrons form covalent bonds with neighboring atoms
- The fifth electron is loosely bound and easily moves to the conduction band
- **Majority carriers:** Electrons
- **Minority carriers:** Holes

3.3 P-Type Semiconductor

- Formed by doping intrinsic semiconductor with **trivalent** impurities (Boron, Gallium, Indium)
- Three valence electrons form covalent bonds, creating a **hole** (missing electron)
- **Majority carriers:** Holes
- **Minority carriers:** Electrons

Topic 4: PN Junction Diode

4.1 Formation of PN Junction

A **PN junction** is formed when a **p-type** semiconductor is joined to an **n-type** semiconductor.

4.2 Depletion Region

When a p-type and n-type semiconductor are joined, the following occurs:

1. **Diffusion:** Electrons from the n-side diffuse into the p-side, and holes from the p-side diffuse into the n-side
2. **Recombination:** Electrons and holes recombine near the junction
3. **Depletion Region:** A region forms around the junction with no free charge carriers
4. **Potential Barrier:** Due to the charge separation, a potential difference develops across the junction:
    - **Silicon:** ~0.7 V
    - **Germanium:** ~0.3 V

```
PN JUNCTION DIAGRAM

P-TYPE             DEPLETION          N-TYPE
                   REGION
+-----------------------------------------------+
|  Holes           - - - + + +      Electrons   |
| (Majority)       - - - + + +     (Majority)   |
|                  - - - + + +                  |
|  Electrons                      Holes         |
| (Minority)                    (Minority)      |
+-----------------------------------------------+
```

Topic 5: Biasing of PN Junction

5.1 Forward Bias

Definition: When the **positive terminal** of a battery is connected to the **p-side** and the **negative terminal** to the **n-side**.

Effect:

1. The external voltage **reduces** the potential barrier
2. The depletion layer becomes **narrower**
3. Electrons easily flow from n-side to p-side
4. A current of **milliamperes** flows through the diode

Forward Bias Characteristics

- Current increases exponentially with voltage
- Small changes in bias voltage cause large changes in current
- The forward resistance is only a few ohms
- Silicon diode: Current starts flowing at ~0.7V
- Germanium diode: Current starts flowing at ~0.3V

```
FORWARD BIAS CONFIGURATION

  P                      N
+--+                    +--+
|  |                    |  |
+--+--+--+   +------+   +--+
   |  |      |      |      |  |
   +--+      |      |      +--+
           + ------- +
           |  1.5V   |
           + ------- +
         (Positive to p-side)
         (Negative to n-side)

     Current flows >>>>>>>
```

5.2 Reverse Bias

Definition: When the **positive terminal** of a battery is connected to the **n-side** and the **negative terminal** to the **p-side**.

Effect:

1. The external voltage **increases** the potential barrier
2. The depletion layer becomes **wider**
3. Current flow is **significantly reduced**
4. Only a **very small reverse current** (microamperes) flows due to minority carriers

Reverse Bias Characteristics

- Very small current flows (leakage current)
- Reverse resistance is very high (mega-ohms)
- At **breakdown voltage**, current suddenly increases sharply

```
REVERSE BIAS CONFIGURATION

  P                     N
+--+                  +--+
|  |                  |  |
+--+--+--+   +-----+   +--+
   |  |     |     |     |  |
   +--+     |     |     +--+
          + ------- +
          |  1.5V   |
          + ------- +
        (Negative to p-side)
        (Positive to n-side)

     Very little current flows
```

Topic 6: Diode Characteristics & Applications

6.1 VI Characteristics of PN Junction

```
VI CHARACTERISTICS OF PN JUNCTION DIODE

  Reverse Bias   |  Forward Bias
                 |
     -V  <-----+----->  +V
                 |
   Reverse       |      Forward
   Current       |      Current
      |          |         |
   +--+          |       +-+-+
   |  |          |       | | |
   +--+          |       +-+-+
      |          |         |
      +---------+---------+--> Voltage
  Breakdown      |   0.7V (Si)
                 |   0.3V (Ge)
```

6.2 Diode as a Switch

Bias Condition	Diode State	Switch Position
Forward Bias	Conducts (ON)	Closed (Switch ON)
Reverse Bias	Blocks (OFF)	Open (Switch OFF)

6.3 Half-Wave Rectifier

A **half-wave rectifier** converts AC to DC by allowing only one half-cycle of the AC signal to pass through. During the positive half-cycle, the diode is forward-biased and conducts. During the negative half-cycle, the diode is reverse-biased and blocks current.

6.4 Full-Wave Rectifier

A **full-wave rectifier** converts both halves of the AC signal to DC. It uses two diodes and a center-tapped transformer (or four diodes in a bridge configuration). Both positive and negative half-cycles are rectified, producing a smoother DC output.

Topic 7: Zener Diode and Voltage Regulation

7.1 Zener Diode

A **Zener diode** is a special type of diode designed to operate in the **reverse breakdown region**. It maintains a constant voltage across its terminals regardless of current changes, making it ideal for **voltage regulation**.

Key Characteristics

- Operates in reverse bias (breakdown region)
- Maintains constant voltage (Zener voltage V_Z)
- Used as a voltage reference/regulator
- Unlike ordinary diodes, Zener diodes are designed to handle breakdown without damage

7.2 Voltage Regulation

A Zener diode can be used as a **voltage regulator** by connecting it in parallel with the load and using a series resistor to limit current. The Zener diode maintains a constant output voltage even when the input voltage or load current varies.

```
ZENER VOLTAGE REGULATOR

    +---Rs---+-------+
    |        |       |
  Vin        Z       RL (Load)
    |        |       |
    +--------+-------+
             GND

Vout = VZ (constant)
```

Topic 8: Bipolar Junction Transistor (BJT)

8.1 Introduction

A **Bipolar Junction Transistor (BJT)** is a three-terminal semiconductor device that can amplify or switch electronic signals.

8.2 BJT Structure

Terminal	Description	Function
Emitter (E)	Heavily doped	Emits charge carriers
Base (B)	Thin, lightly doped	Controls current flow
Collector (C)	Moderately doped	Collects charge carriers

```
BJT STRUCTURE

NPN Transistor:         PNP Transistor:
Emitter (N) -> Base (P) -> Collector (N)
                        Emitter (P) -> Base (N) -> Collector (P)

  Collector (C)             Collector (C)
      |                          |
  +---+---+                  +---+---+
  | N     |                  | P     |
  +--------+                 +--------+
  | P     |   Base (B)       | N     |   Base (B)
  +--------+                 +--------+
  | N     |                  | P     |
  +---+---+                  +---+---+
      |                          |
  Emitter (E)                Emitter (E)
```

8.3 Transistor Configurations

Configuration	Common Terminal	Input	Output	Current Gain	Voltage Gain

Common Base (CB)	Base	Emitter	Collector	< 1	High
Common Emitter (CE)	Emitter	Base	Collector	> 1	High
Common Collector (CC)	Collector	Base	Emitter	~1	< 1

Common Emitter (CE) configuration is the most widely used configuration for amplification and switching.

8.4 BJT as a Switch (CE Configuration)

1. Cut-off Region (Switch OFF):

- Base current (I_B) = 0
- No current flows from collector to emitter
- Transistor acts as an **open switch**

2. Saturation Region (Switch ON):

- Base current is sufficient to fully turn on the transistor
- Maximum current flows from collector to emitter
- Transistor acts as a **closed switch**

```
TRANSISTOR AS A SWITCH - CE CONFIGURATION

               Vcc
                 |
               +-+-+
               | Rc |
               +-+-+
                 |
                 +----+----> Vout
                 |
     Base ---+---+---+
     (B)     |   C   |
             +---+---+
             |   E   |
             +---+---+
```

```
         |
        GND
```

8.5 Saturated and Non-Saturated Logic

Type	Description	Examples
Saturated Logic	Transistors operate in saturation region, faster switching times	TTL (Transistor-Transistor Logic)
Non-Saturated Logic	Transistors operate in active region, lower power consumption	ECL (Emitter-Coupled Logic)

Topic 9: Integrated Circuits (ICs)

9.1 What are Integrated Circuits?

An **Integrated Circuit (IC)** is a **miniature electronic circuit** consisting of thousands or millions of semiconductor devices (transistors, diodes, resistors) fabricated on a single piece of silicon.

9.2 Advantages of ICs

Advantage	Description
Small Size	Millions of components on a tiny chip
Low Cost	Mass production reduces cost
Low Power	Less power consumption
Reliability	Fewer connections, fewer failures
Speed	Faster operation
No Wiring	No external connections needed

Topic 10: Characteristics of Digital Logic Families

10.1 TTL (Transistor-Transistor Logic)

Parameter	TTL
Supply Voltage	5V
Logic 0 (Input)	0V to 0.8V
Logic 1 (Input)	2V to 5V
Propagation Delay	10-20 ns
Power Dissipation	Moderate
Fan-out	Up to 10
Noise Margin	Good

Advantages: Reliable, easy to use, wide range of available ICs.

Disadvantages: High power consumption compared to CMOS, less noise margin than CMOS.

10.2 ECL (Emitter-Coupled Logic)

Parameter	ECL
Supply Voltage	-5.2V
Propagation Delay	1-2 ns (fastest)
Power Dissipation	Very high

Fan-out	Up to 25
Noise Margin	Very poor

Advantages: Very fast switching, good for high-speed systems.

Disadvantages: High power consumption, complex power supply requirements, poor noise margin.

10.3 CMOS (Complementary Metal-Oxide-Semiconductor)

Parameter	CMOS
Supply Voltage	3V to 18V
Propagation Delay	10-50 ns
Power Dissipation	Very low
Fan-out	Up to 50
Noise Margin	Very good

Advantages: Very low power consumption, high noise immunity, wide voltage range, high fan-out capability.

Disadvantages: Slower than TTL, ESD sensitive.

10.4 Comparison of Logic Families

Parameter	TTL	ECL	CMOS
Speed	Medium	Fastest	Slowest
Power	Moderate	Very High	Very Low
Noise Margin	Good	Poor	Very Good
Fan-out	10	25	50

Cost	Low	High	Low
Voltage	5V	-5.2V	3-18V

Questions

Short Answer Questions (2-5 Marks)

1. What is an energy band diagram?
2. What is a PN junction diode?
3. What is forward bias and reverse bias?
4. What is depletion region?
5. What is a BJT? Name its terminals.
6. What are the types of BJT configurations?
7. What is TTL logic family?
8. What is CMOS logic family?
9. What is an integrated circuit?
10. What is the difference between forward and reverse bias?

Long Answer Questions (10 Marks)

1. Explain the energy band diagram of insulators, semiconductors, and conductors.
2. Explain the formation of PN junction and depletion region.
3. Explain forward and reverse bias of a PN junction with diagrams.
4. Explain the working of a BJT as a switch.
5. Compare and contrast TTL, ECL, and CMOS logic families.

HPU Exam-Based Questions

1. State De Morgan's theorem.
2. Compare EEPROM and flash memory.
3. State the difference between Fan-in and Fan-out.

Objective Questions (1 Mark)

1. The forbidden band gap of silicon is approximately:
   a. 0.3 eV
   b. 0.7 eV

   c. 1.1 eV
   d. 5 eV

Answer: (c) 1.1 eV

2. Which logic family has the highest speed?
   a. TTL
   b. ECL
   c. CMOS
   d. None

Answer: (b) ECL

3. Which logic family has the lowest power consumption?
   a. TTL
   b. ECL
   c. CMOS
   d. None

Answer: (c) CMOS

4. The majority carriers in N-type semiconductor are:
   a. Holes
   b. Electrons
   c. Protons
   d. Neutrons

Answer: (b) Electrons

5. The potential barrier for silicon diode is approximately:
   a. 0.3V
   b. 0.7V
   c. 1.1V
   d. 5V

Answer: (b) 0.7V

Quick Revision

PN Junction Summary

```
PN Junction
   |
   +--- Forward Bias --- p-side positive, n-side negative
   |                     --- Reduced potential barrier
   |                     --- Current flows (mA)
   |                     --- Diode ON
   |
   +--- Reverse Bias --- n-side positive, p-side negative
                         --- Increased potential barrier
                         --- Very small current flows (uA)
                         --- Diode OFF
```

Logic Family Comparison

Parameter	TTL	ECL	CMOS
Speed	Medium	Fastest	Slowest
Power	Moderate	Very High	Very Low
Noise Margin	Good	Poor	Very Good

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